Hardware security and trust

Computer Assignment 1

Due date : 1404/01/13

Question 2

Lightweight RFID Mutual Authentication Protocol

In this assignment, you will implement a simplified version of the **Lightweight RFID Mutual Authentication Protocol**. This protocol ensures secure communication between an RFID tag (client) and an RFID reader (server) through **mutual authentication**. You will utilize a **Pre-Shared Key (PSK)**, **random numbers (R1, R2)**, and **hash functions (SHA-0 or SHA-1, depending on your student ID)** to verify integrity and authentication. Before starting, carefully read the paper [1]. This paper introduces a lightweight authentication mechanism designed for RFID systems, balancing security and computational efficiency. Due to the limited resources of RFID tags, traditional encryption methods (e.g., RSA, AES) are infeasible. Instead, lightweight operations such as hashing, XOR masking, and pseudo-random number generation are employed. Your task is to implement and evaluate this protocol.

Assignment Tasks

1. Setting Up the Environment and Protocol Implementation (30 marks)

Implement two components: **Reader (server)** and **RFID Tag (client)**. The Reader initiates authentication, while the Tag responds to authentication requests.

Pre-Shared Key (PSK):

Use your **Student ID** as the PSK for mutual authentication.

Hash Function:

Use **SHA-256** if your Student ID is even, or **SHA-512** if odd.

Python Scripts:

Create two scripts: **server.py** and **client.py**. The **server.py** script implements the Reader, while the **client.py** script implements the Tag. Simulate communication between the two to implement the protocol.

Guidance for Communication:

Use Pythons **`socket`** library to establish communication between the Reader and Tag. The **Reader (server)** should act as the communication initiator by creating a socket, binding it to an IP and port, and listening for incoming connections. The **Tag (client)** should connect to the Reader by creating a socket and connecting to the Readers IP and port. Use the **`send()`** and **`recv()`** methods to exchange messages between the Reader and Tag.

2. Functional Testing (20 marks)

Success Scenario:

Demonstrate successful mutual authentication using the correct PSK. Log steps in **server.log** for Reader logs and **client.log** for Tag logs. Include random numbers (R_1, R_2), sent/received messages, hash values (H_1, H_2), and authentication results.

Failure Scenarios:

Test the following failure scenarios: **Wrong PSK**, where modifying the PSK shows authentication failure; **Replay Attack**, where reusing a previous session demonstrates failure due to fresh random values; and **Man-in-the-Middle Attack**, where modifying a message in transit shows how the protocol detects it.

This question carries **50 marks**, and along with **50 marks** from Question 1, the total marks for the entire assignment sum up to **100 marks**.
For any questions or clarifications, please contact **mehrabpsh@gmail.com** or send a DM in Telegram through the ID @Ciriaa11.
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References

[1] S. Azad and B. Ray, "A lightweight protocol for rfid authentication," in *2019 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE)*, pp.1–6, 2019.